Lecture Plan

B.Tech. CSE-Cyber Security

Subject Name: IT Fundamentals 30 hrs +10 experiments

Subject Code:

Sl. No.	Description	Bloom's level*
CO1	Identify and install computer hardware components, including motherboards, storage devices, and memory.	Remember, Understand
CO2	Troubleshoot common PC hardware, networking, and OS issues.	Apply, Analyze
CO3	Configure operating systems, including Windows, Linux, and Mac OS.	Apply, Analyze
CO4	Implement security measures for systems, networks, and mobile devices.	Evaluate
CO5	Apply virtualization and cloud computing concepts for IT infrastructure management.	Evaluate

S.No.	Topic	Lecture No.
1.	**Unit 1 – Computer Hardware and Troubleshooting** **Introduction to Computer Hardware** • Components of a computer system • Motherboards and chipsets • CPUs and processor architecture • Memory (RAM & ROM) fundamentals	**Lecture 1**
2.	**Storage Devices and System Assembly** • HDD, SSD, NVMe drives • Installing storage devices • BIOS/UEFI overview • Boot sequence	**Lecture 2**
3.	**Power Supply and Computer Components** • SMPS and power ratings • Connectors and cables • Expansion cards • Cooling systems	**Lecture 3**
4.	**Hardware Installation and Troubleshooting** • Hardware installation procedures • Power, display, and POST errors • BIOS troubleshooting • Preventive maintenance	**Lecture 4**
5.	**Hardware Diagnostics and Practical Lab** • Hardware diagnostic tools	**Lecture 5**

	• Cable testing • Memory diagnostics • Unit revision	
6.	**Unit 2 – Networking and Virtualization Networking Fundamentals** • Types of computer networks • Network topologies • Networking hardware • Communication models	**Lecture 6**
7.	**Network Devices and Cabling** • Switches, Routers, Access Points • Ethernet standards • UTP, STP, Fiber optics • Connectors	**Lecture 7**
8.	**Wireless Networking** • Wi-Fi standards • Wireless security protocols • SSID and Channels • Troubleshooting wireless issues	**Lecture 8**
9.	**Virtualization Fundamentals**• Virtual Machines• Hypervisors• VirtualBox and VMware• Benefits of virtualization	**Lecture 9**
10.	**Cloud Computing Concepts**• Cloud service models• Deployment models• Virtualization vs Cloud• Unit revision	**Lecture 10**
11.	**Unit 3 – Mobile and Print Device SupportMobile Device Fundamentals**• Mobile operating systems• Smartphone components• Tablets and laptops	**Lecture 11**
12.	**Mobile Device Configuration**• Device setup• Synchronization• Email configuration• Connectivity options	**Lecture 12**
13.	**Troubleshooting Mobile Devices**• Common hardware issues• Software issues• Battery troubleshooting• Mobile security basics	**Lecture 13**
14.	**Printers and Peripheral Devices**• Printer technologies• Installing printers• Printer configuration• Common printer problems	**Lecture 14**
15.	**Windows User and System Configuration**• User accounts• Control Panel• System settings• Unit revision	**Lecture 15**
16.	**Unit 4 – Operating Systems and SecurityOperating System Fundamentals**• Functions of an OS• Windows architecture• Installation process• Boot management	**Lecture 16**
17.	**Windows Administration**• Device Manager• Task Manager• Disk Management• System utilities	**Lecture 17**

18.	**Linux and macOS Fundamentals**• Linux overview• macOS overview• Comparison of Windows, Linux and macOS	**Lecture 18**
19.	**Windows Security**• User permissions• Windows Defender• Firewall configuration• Security policies	**Lecture 19**
20.	**Operating System Troubleshooting**• Boot failures• Recovery options• System Restore• Unit revision	**Lecture 20**
21.	**Unit 5 – SOHO Networks and Operational ProceduresSOHO Networking Fundamentals**• Home and small office networks• Router configuration• Internet connectivity	**Lecture 21**
22.	**SOHO Router Security**• Wireless security• Password policies• Firmware updates• Access control	**Lecture 22**
23.	**Mobile OS Security**• Device encryption• Authentication methods• Application permissions• Security best practices	**Lecture 23**
24.	**Troubleshooting Security Issues**• Malware protection• Security updates• Common threats• Incident reporting	**Lecture 24**
25.	**Operational Procedures**• Documentation standards• Communication skills• Professional ethics• Workplace safety	**Lecture 25**
26.	**Integrated Hardware and Networking Lab**• Hardware assembly• Network configuration• Practical troubleshooting	**Lecture 26**
27.	**Integrated Operating System and Security Lab**• Windows administration• User management• Firewall configuration	**Lecture 27**
28.	**Integrated Mobile, Printer and SOHO Lab**• Device setup• Printer installation• Router configuration	**Lecture 28**
29.	**Course Revision and Problem Solving**• Comprehensive revision• Scenario-based troubleshooting• Question discussion	**Lecture 29**
30.	**Final Review and Assessment Preparation**• Mock assessment• Practical review• Course summary and feedback	**Lecture 30**

List of Experiments

Experiment No.	Practical Experiment
Experiment 1	**Identification of Computer Hardware Components:** Identify and inspect the motherboard, CPU, RAM, HDD/SSD, SMPS, expansion cards, ports, cables, and peripheral devices available in the laboratory system.
Experiment 2	**System Information and Hardware Diagnostics:** Use Windows System Information (MSINFO32), Device Manager, Task Manager, and Command Prompt to examine hardware specifications, storage devices, memory usage, processor details, and installed drivers.
Experiment 3	**BIOS/UEFI Configuration:** Access BIOS/UEFI settings, verify hardware configuration, modify boot priority, enable virtualization support, and observe BIOS security settings.
Experiment 4	**Operating System User and File Management:** Create multiple user accounts, configure passwords, assign user permissions, create folders with different access rights, and manage files using Windows File Explorer and Command Prompt.
Experiment 5	**Basic Network Configuration and Troubleshooting:** Configure static and dynamic IP addresses, verify network connectivity using **ipconfig**, **ping**, **tracert**, **nslookup**, and troubleshoot basic LAN connectivity issues.
Experiment 6	**Wireless Network and SOHO Router Configuration:** Connect systems to a Wi-Fi network, access the router management interface, configure SSID, change administrator credentials, enable WPA2/WPA3 security, and verify Internet connectivity.
Experiment 7	**Printer Installation and Configuration:** Install a local or network printer, configure printer sharing, print a test page, manage the print queue, and troubleshoot common printing problems.
Experiment 8	**Virtual Machine Installation:** Install Oracle VirtualBox, create a virtual machine, install Ubuntu Linux, configure virtual networking, and perform basic Linux operations.
Experiment 9	**Windows Security Configuration:** Configure Windows Firewall, perform Windows Update, enable Microsoft Defender Antivirus, configure password policies, and review Event Viewer logs for security events.
Experiment 10	**System Maintenance and Preventive Troubleshooting:** Perform Disk Cleanup, Disk Management, CHKDSK, storage optimization, system backup and restore, uninstall/install software, and document the troubleshooting process for common hardware and software issues.

Suggested Readings and References

Sr. No.	Author	Name of the Book	Publisher	Year of Publication
1	Jean Andrews	A+ Guide to IT Technical Support	Cengage Learning	2021
2	Charles Brooks	CompTIA A+ Certification Study Guide	McGraw Hill	2019
3	Mike Meyers	CompTIA A+ Certification All-in-One Guide	McGraw Hill	2022
4	Andrew S. Tanenbaum	Computer Networks	Pearson	2019
5	William Stallings	Operating Systems: Internals and Design Principles	Pearson	2021